

3.10 Shear Force And Bending Moment Diagrams For Over-Hanging Beams

If the end portion of a beam is extended beyond the support such beam is known as overhanging beam. In case of overhanging beams, the B.M. is positive between the two supports, whereas the B.M. is negative for the over-hanging portion. Hence at some point, the B.M. is zero after changing its sign from positive to negative or vice-versa. That point is known as the point of contraflexure or point of inflexion.

3.10.1 Beam With Both Sides Overhang: Point Loads

Let us first calculate the reactions, by taking moments about A or about B.

Let us take moments of the forces on the beam about B.

$$8 \times 5 = R_A \times 4 + 10 \times 2$$

$$\therefore R_A = 5 \text{ kN}$$

$$\therefore R_B = 8 + 10 - 5 = 13 \text{ kN}$$

S.F.D.: For the section CA,

$$F_x = - 8 \text{ kN (Constant)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{Section AB,} \quad F_x = - 8 + 5 = - 3 \text{ kN (Constant)}$$

$$\text{Section BD,} \quad F_x = - 8 + 5 + 13 = +10 \text{ kN}$$

Thus S.F. changes sign at B.

The shear force diagram is shown in Fig 3.14 (b).

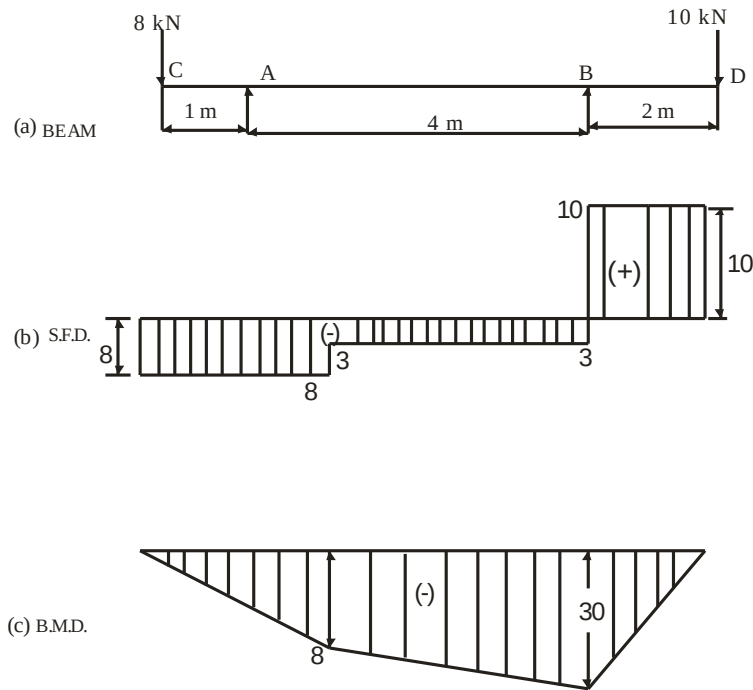


Fig. 3.14

B.M.D.

Section CA, $M_x = -8 \cdot x$ (Linear)

$$\text{At C, } x = 0 \quad \therefore M_C = -8 \times 0 = 0$$

$$\text{At A, } x = 1 \quad \therefore M_C = -8 \times 1 = -8 \text{ kN-m}$$

Section AB, $M_x = -8 \cdot x + 5 \cdot (x-1)$ (Linear)

$$\text{At A, } x = 1 \quad \therefore M_A = -8 \text{ kN-m}$$

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

Section BD, $M_x = -8 \cdot x + 5 \cdot (x-1) + 13 \cdot (x-5)$ (Linear)

$$\text{At B, } x = 5 \quad \therefore M_B = -20 \text{ kN-m}$$

$$\text{At D, } x = 7 \quad \therefore M_D = 0$$

The B.M. diagram is drawn in Fig. 3.14 (c). Maximum bending moment is where S.F. changes sign.

3.10.2 Beam With Both Sides Overhang: Uniformly Distributed Load

Fig. 3.15 (a) shows a beam EABD of length $(l+2a)$ with supports at A and B. The beam carries a udl w N/m over entire length.

Since the beam is symmetrically loaded the reactions R_A and R_B are

$$R_A = R_B = \frac{w}{2}(l + 2a)$$

S.F.D.: For the section EA,

$$F_x = -wx \text{ (Linear)}$$

The shear force will be negative as the resultant force on the left portion of the section is acting downwards.

$$\text{S.F. at E, } x = 0 \therefore F_E = 0$$

$$\text{S.F. at A, } x = a \therefore F_A = -w.a$$

For the section AB,

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at A, } x = a \therefore F_A = + \frac{wl}{2} \text{ (Sudden change in S.F. due to point load } R_A)$$

$$F_x = R_A - wx = \frac{w}{2}(l + 2a) - wx \text{ (Linear)}$$

$$\text{S.F. at B, } x = (a+l) \therefore F_B = - \frac{wl}{2}$$

Hence from A to B the S.F. changes uniformly from $+\frac{wl}{2}$ to $-\frac{wl}{2}$. The S.F. at C the middle point of the beam is zero.

For the section BD, the S.F. will change uniformly from $+wa$ at B to zero at D.

The shear force diagram is shown in Fig 3.15 (b).

B.M.D.

Section EA, $M_x = -\frac{wx^2}{2}$ (Hogging and parabolic)

$$\text{At E, } x = 0 \quad \therefore M_E = 0$$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

$$\text{Section AB, } M_x = R_a(x-a) - \frac{wx^2}{2} = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$$

$$\text{At A, } x = a \quad \therefore M_A = -\frac{wa^2}{2}$$

$$\text{At C, } x = a + \frac{l}{2} \quad \therefore M_C = \frac{w(l^2 - 4a^2)}{8}$$

Case (A): when $l > 2a$

For this case M_c is positive.

For this case there will be two points of contraflexure C_1 and C_2 between A and B. The positions of these points can be determined by equating the general expression for B.M. for any section in AB to zero.

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2}$$

For the points of contraflexure

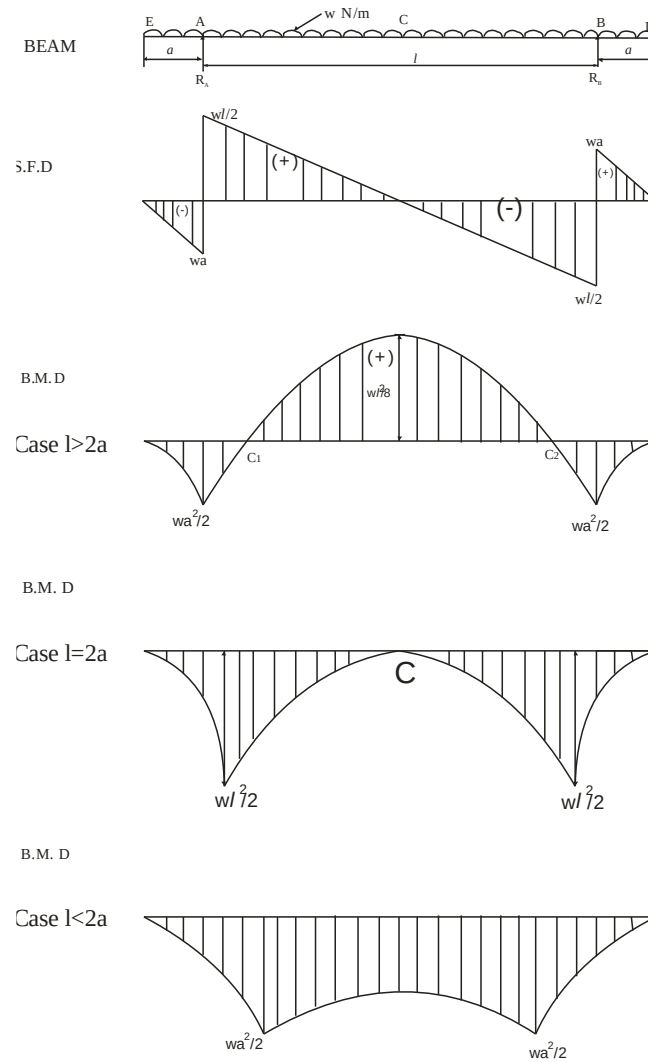


Fig. 3.15

$$M_x = \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } \frac{w}{2}(l+2a)(x-a) - \frac{wx^2}{2} = 0$$

$$\text{Or } x^2 - (2a+l)x + a(2a+l) = 0$$

$$x = \frac{(2a + l) \pm \sqrt{(2a + l)^2 - 4a(2a + l)}}{2}$$

$$\text{or } x = a + \frac{l}{2} \pm \frac{\sqrt{(l)^2 - 4a^2}}{2}$$

$$\text{Here EC} = a + \frac{l}{2}$$

Hence the points of contraflexure C_1 and C_2 are at a distance $\frac{\sqrt{(l)^2 - 4a^2}}{2}$ from the middle point C of the beam. The BMD is as shown in Fig. 3.15 (c).

Case (B): when $l=2a$ B.M. at C = 0

The points of contraflexure will coincide with C and the BMD between A and B just touches the span. The BMD is as shown in Fig. 3.15 (d).

Case (C): when $l < 2a$

$$M_C = \frac{w(l^2 - 4a^2)}{8} \quad \text{since } 4a^2 > l^2 \therefore M_C \text{ is negative.}$$

The BMD is as shown in Fig. 3.15 (e).